Spring 2016

CS3230 Tutorial (Transform and Conquer Algorithms, week of 7 March)

1. Given an array *A[0..n-1]* and a value *V*, design a better than quadratic time efficiency algorithm to determine if there exist $i \neq j$ such that $A[i] + A[j] = V$.

2. Consider a set of *n* open intervals

$$(a_i, b_i) = \{X \in R: a_i < x < b_i\} \; ; \; i = 1,\ldots,n;$$

   where *R* is the set of real numbers. Design a better than quadratic efficiency algorithm to find the maximum number of overlapping intervals. Intervals overlap if their intersection is non-empty.

3. Draw a height 5 AVL tree with the minimal number of nodes.

4. Design and analyze (time efficiency) an algorithm for deleting from a heap respectively (1) the smallest element, (2) the largest element, and (3) the element with a given key.

5. An $n \times n$ matrix of lights is controlled by an $n \times n$ matrix of switches. Switch s_{ij} toggles lights L_{uv} where

$$(u, v) \in \{(x, y): |x - i| + |y - j| \leq 1\}.$$

   Given an initial lighting pattern, describe an algorithm to transform it to another lighting pattern. For example, the following flipping of switches turns off an "identity matrix" of 2×2 lights:

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} + S_{11} \rightarrow \begin{bmatrix} 0 & 1 \\ 1 & 1 \end{bmatrix} + S_{22} \rightarrow \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix};$$

   the following flipping of switches turns on an "identity matrix" of 2×2 lights:

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} + S_{12} \rightarrow \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} + S_{21} \rightarrow \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}.$$

6. Design an algorithm to find a^n based on the Horner evaluation of the ternary representation of *n*.

7. Design an algorithm to solve the farmer, cabbage, goat, wolf, river crossing problem?